

Data Link Control Protocols

- **Flow Control**

Stop-and-Wait Flow Control
Sliding Window Flow Control

- **Error Control**

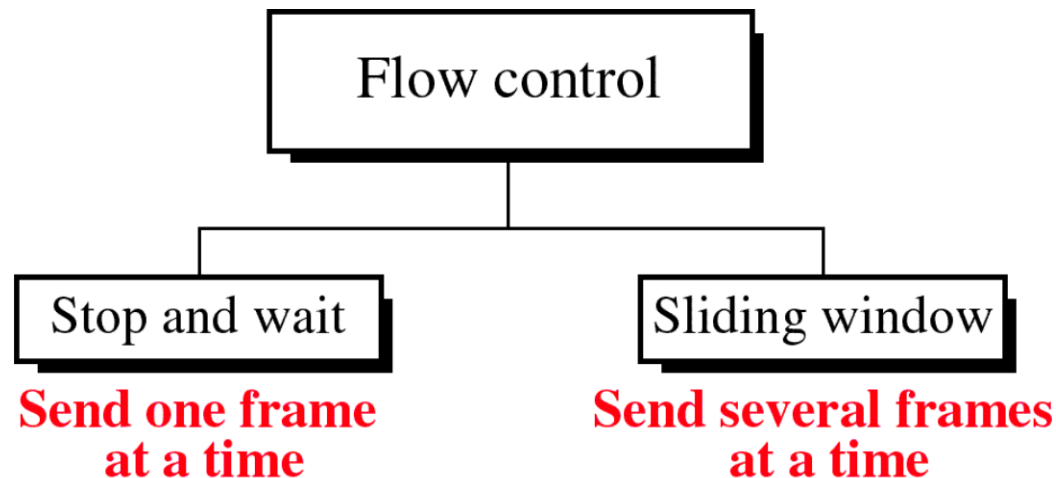
Stop and Wait ARQ
Go-Back-N ARQ
Selective-Reject ARQ

- **High-Level Data Link Control Protocol**

Flow Control

Flow control refers to set of procedures used to restrict the amount of data the sender can send before waiting for acknowledgement.

Each receiving device has a block of memory, called a buffer, reserved for storing incoming data frames until they are processed. If the buffer begins to fill up, the receiver must be able to tell the sender to halt transmission until it is once it is again able to receive.

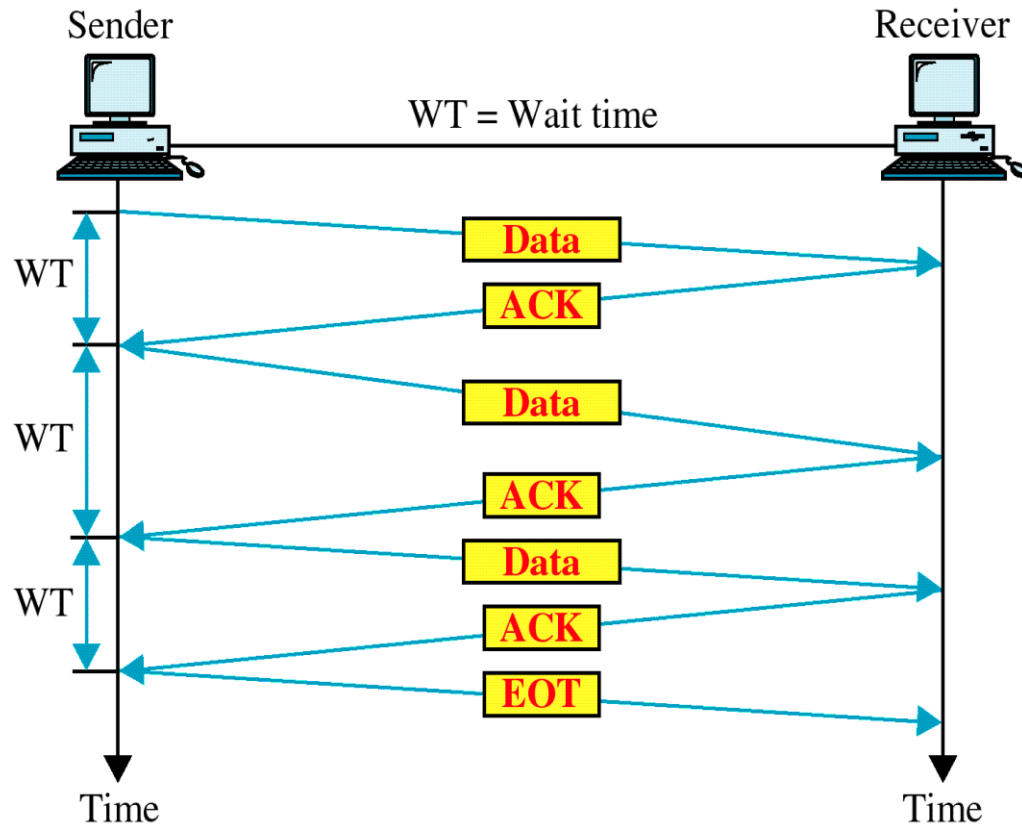


Stop and Wait Flow Control

The sender sends one frame and waits for an acknowledgement before sending the next frame.

The advantage of stop and wait is simplicity.

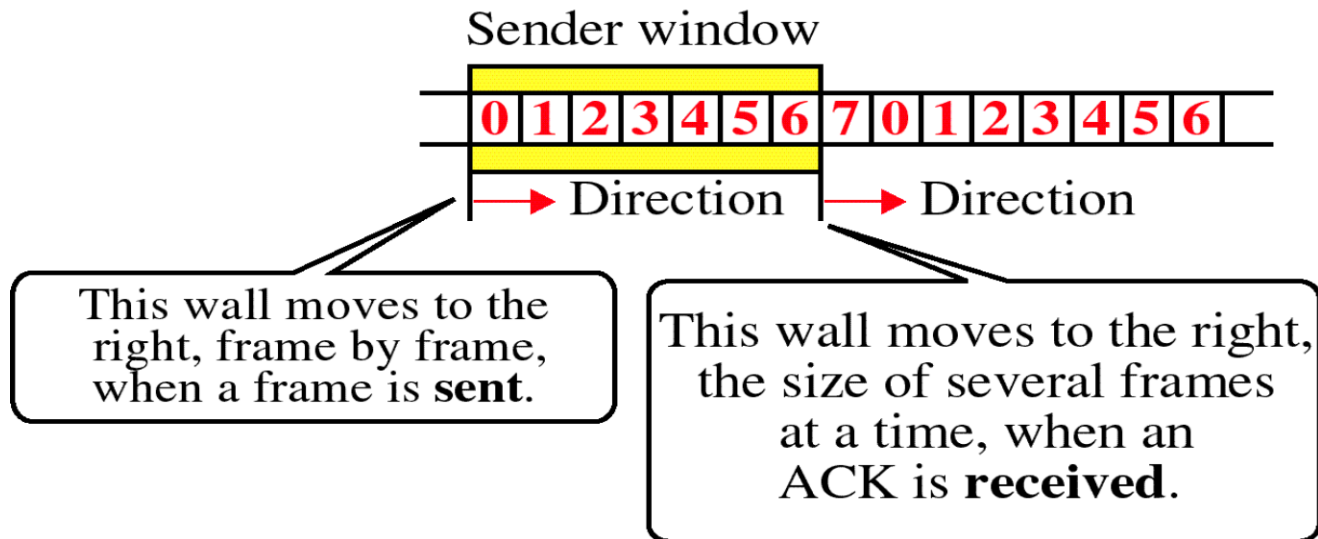
The disadvantage is inefficient. If distance between the sender and receiver is long - waiting time is increases and low utilization of link.



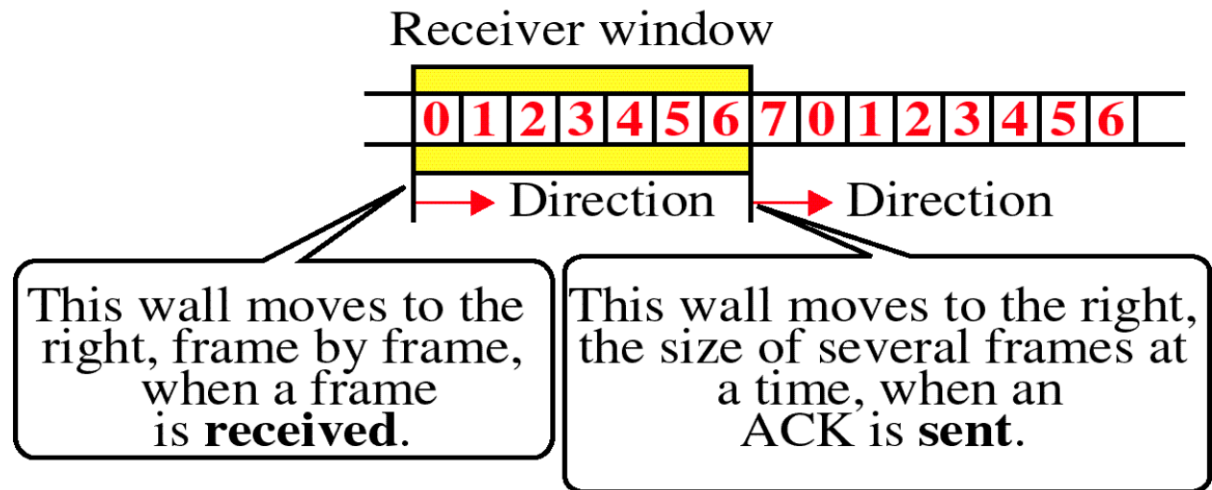
Sliding Window Flow Control

- Both sender and receiver maintain windows (buffers) to hold frames.
- The sender can transmit several frames before needing an acknowledgement.
- The receiver acknowledges only some of the frames, using a single ACK to confirm the receipt of multiple data frames.
- To keep track of which frames have been transmitted and which received, sliding window introduces an identification scheme based on the size of window.
- The frames are numbered (sequence number) modulo- n , which means they are numbered from 0 to $n-1$. The size of the window is $n-1$.
- If $n=8$, the frames are numbered 0, 1, 2, 3, 4, 5, 6, 7, 0, 1, 2, 3, The size of the window is 7.

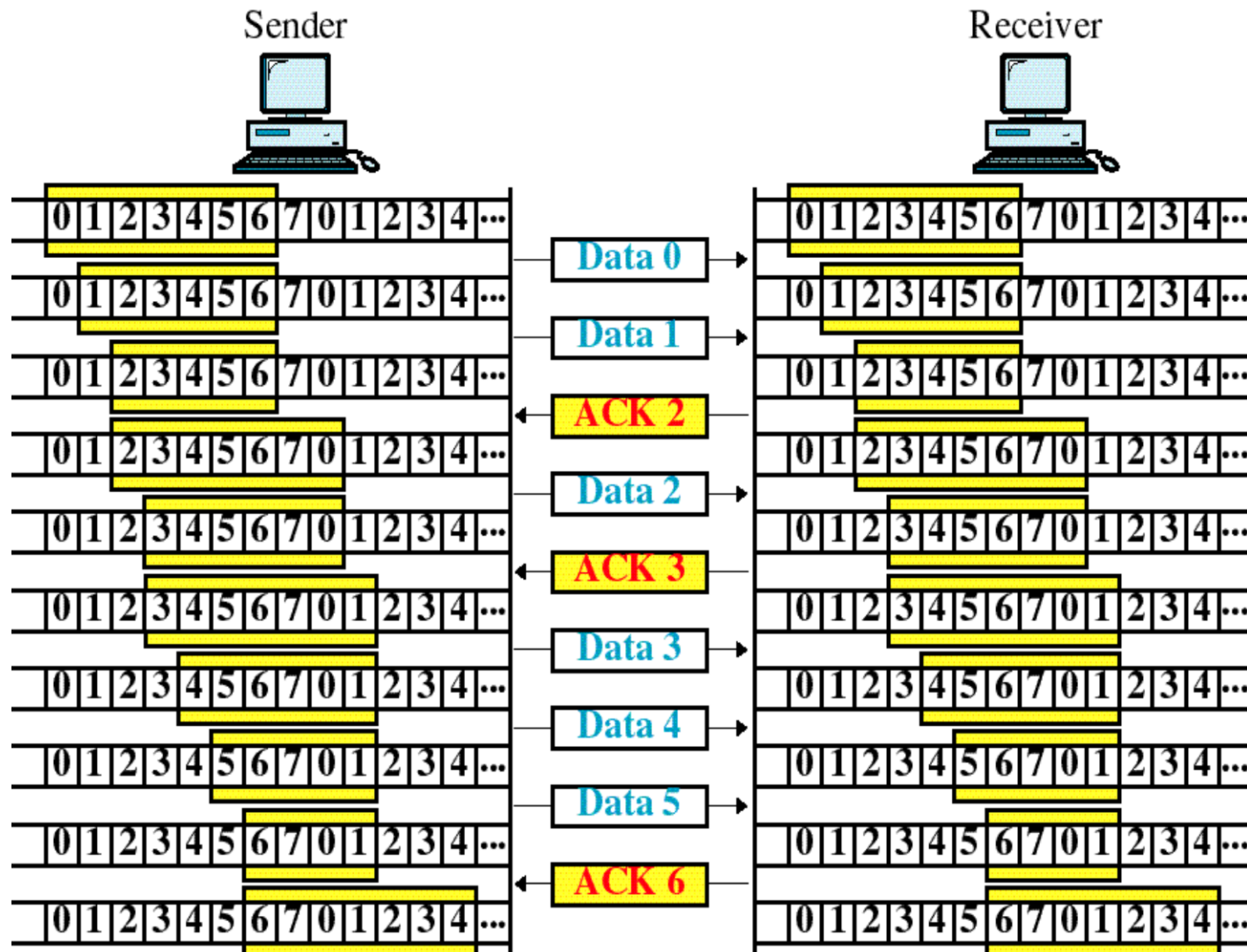
Sender Sliding Window



Receiver Sliding Window



Sliding Window Example



More about window size

The size of window is one less than the modulo range so that there is no ambiguity in the acknowledgement of the received frames.

Assume that the frame sequence numbers are modulo-8 and the window size is also 8.

Now frame 0 is sent and ACK1 is received.

The sender expands its window and sends frames 1, 2, 3, 4, 5, 6, 7, and 0.

If sender receives an ACK1 again. It is now not sure if this is a duplicate of the previous ACK1 (duplicate by network or new ACK1 confirming the most recently sent 1, 2, 3, 4, 5, 6, 7, 0 frames).

Piggybacking

If two stations exchange data, each needs to maintain two windows, one for transmit and one for receive, and each side needs to send the data and acknowledgments to the other. To provide efficient support of this requirement, a feature known as piggybacking.

Data and acknowledgement are sent in the same frame is called piggybacking.

Error Control

Error control in data link layer based on the method of error detection and retransmission.

Automatic Repeat Request (ARQ):

Anytime an error is detected during transit, a negative acknowledgement (NAK) is returned and the specified frames are retransmitted. This process is called ARQ.

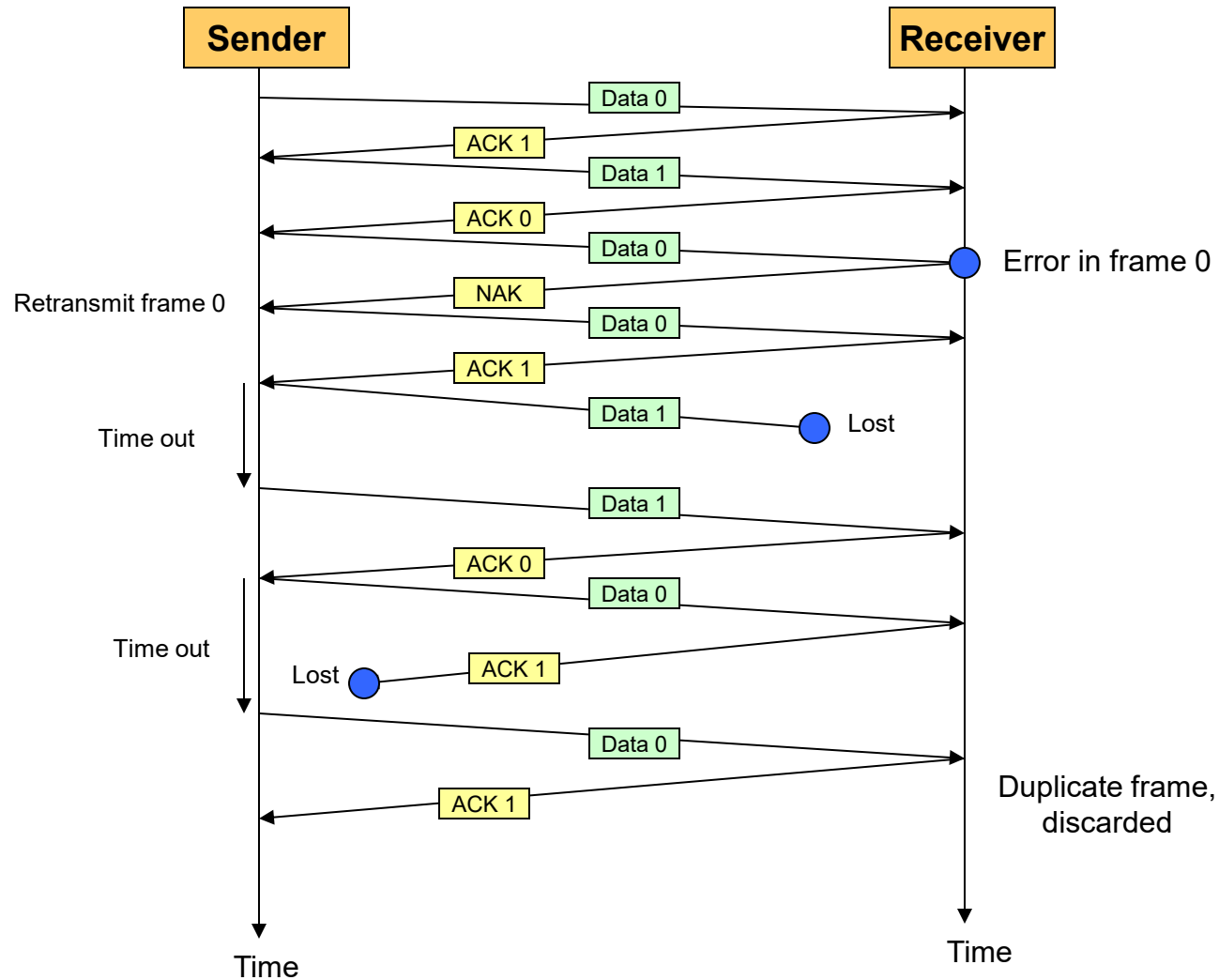
Retransmission of data frame in the three cases: damaged frame, lost frame and lost acknowledgment.

Stop and Wait ARQ:

Stop and wait ARQ is a form of the stop and wait flow control extended to include retransmission of data in case of lost or damaged frames. For retransmission

- sender keeps a copy of the last frame transmitted until it receives an acknowledgment for that frame.
- both data and ACK frames are numbered alternately 0 and 1.
- A NAK is returned when frame has been corrupted.
- sender is equipped with a timer.

Stop and Wait ARQ, damaged frame, lost frame and lost acknowledgement:



Sliding Window ARQ

There are two types of sliding window ARQ. Go-back-N ARQ and Selective-reject ARQ. Three features are added to the basic sliding window flow control to handle the errors.

- The sender keeps copies of all transmitted frames until they have been acknowledged.
- A NAK is returned when a frame has been corrupted.
- Sender equipped with a timer.

Go-back-N ARQ:

If one frame is lost or damaged, all frames sent since the last frame acknowledgement are transmitted.

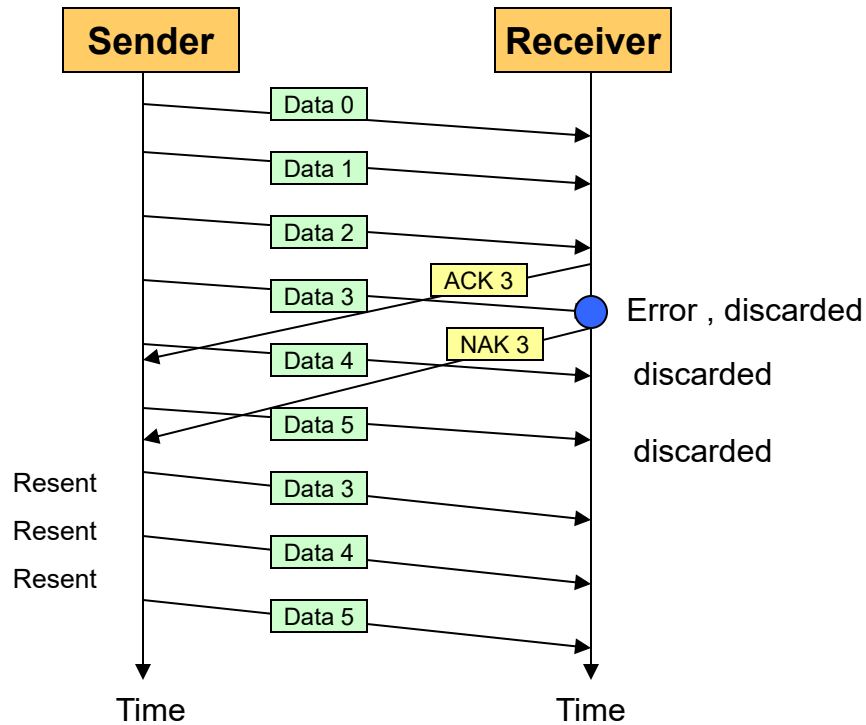
Selective-reject ARQ:

Only a specified damaged or lost frame is retransmitted if a frame is corrupted in transit, a NAK is returned and the frame is resent out of sequence.

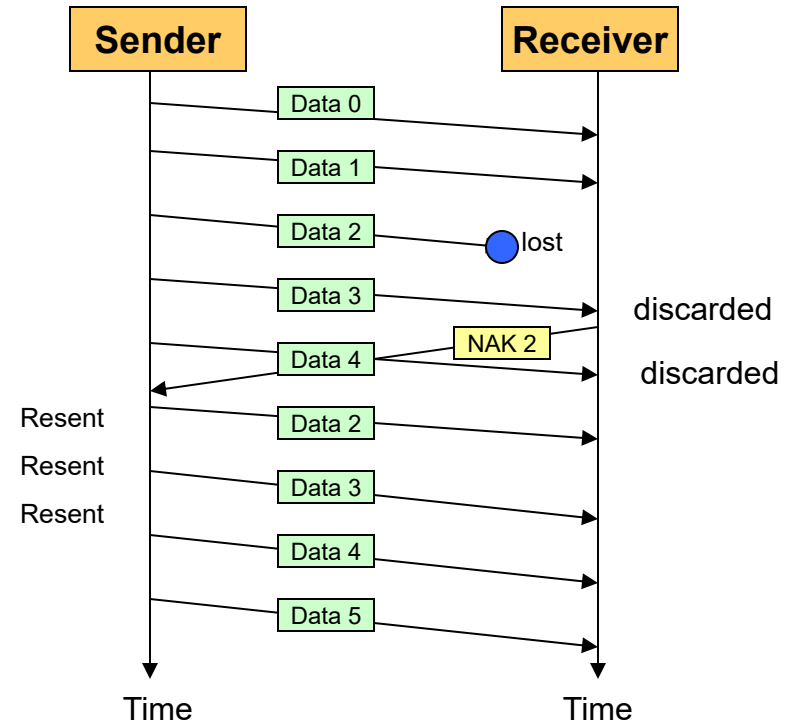
- Also called selective retransmission
- Only rejected frames are retransmitted
- Subsequent frames are accepted by the receiver and buffered
- Minimizes retransmission
- Receiver must maintain large enough buffer

Go-back-N ARQ

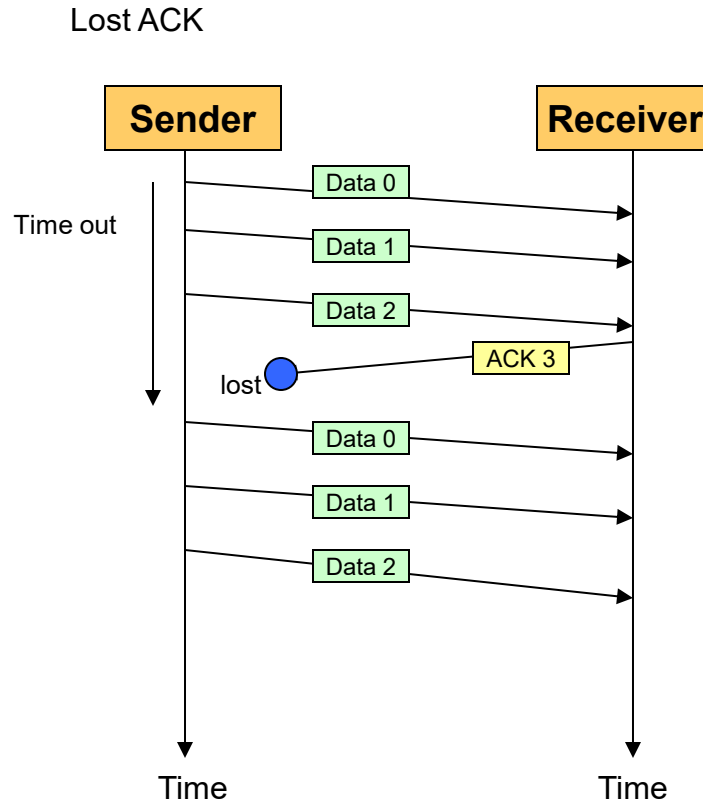
Damaged data frame



Lost data frame



Go-back-N ARQ

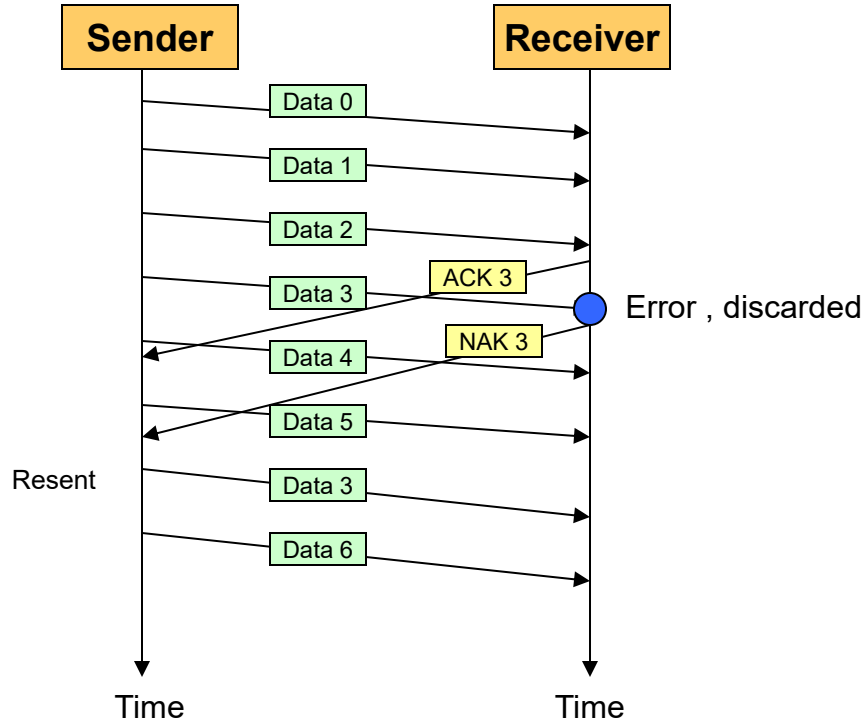


Selective-reject ARQ:

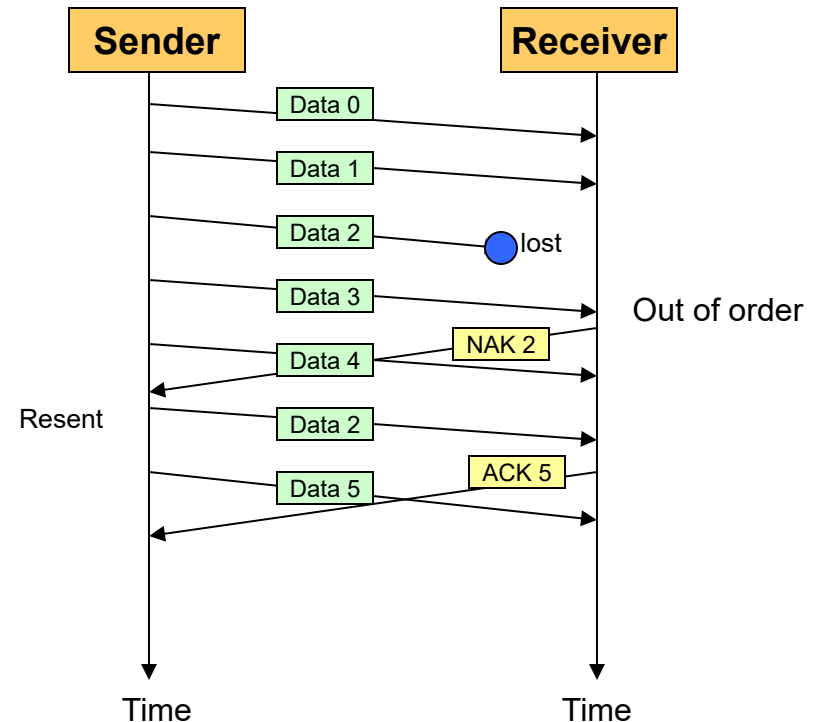
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Damaged data frame



Lost data frame



Lost Acknowledgement:

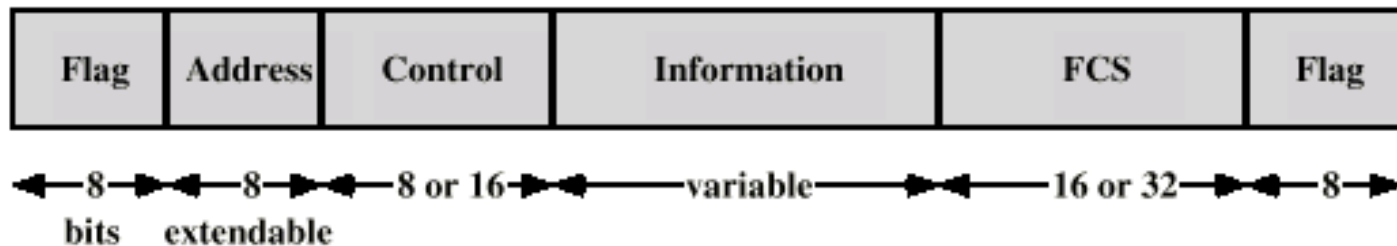
Lost ACK or NAK frames are treated as they by go-back-n ARQ.

High Level Data Link Control Protocol

- Adapted as a part of x.25 network interface.
- It is bit oriented protocol.
- Its uses bit stuffing for data transparency.

Frame Structure:

- Synchronous transmission
- All transmissions in frames
- Single frame format for all data and control exchanges



(a) Frame format

Flag Fields

- Delimit frame at both ends
- 01111110
- Receiver hunts for flag sequence to synchronize
- Bit stuffing used to avoid confusion with data containing 01111110
 - 0 inserted after every sequence of five 1s
 - If receiver detects five 1s it checks next bit
 - If 0, it is deleted
 - If 1 and seventh bit is 0, accept as flag
 - If sixth and seventh bits 1, sender is indicating abort

Address Fields: identify the terminals.

Information Field: may contain any information.

Checksum: uses for CRC.

Control Fields:

- Uses for sequence number, acknowledgment and other purposes.
- There are three types of frames.
 - **Information** - data to be transmitted to user (next layer up)
 - Flow and error control piggybacked on information frames
 - **Supervisory** - ARQ when piggyback not used
 - **Unnumbered** - supplementary link control. It carry data when unreliable connectionless service is called for.